C

ORDINARY DIFFERENTIAL EQUATIONS • UNIT 5.01

Variation of Parameters & Greens Kernel

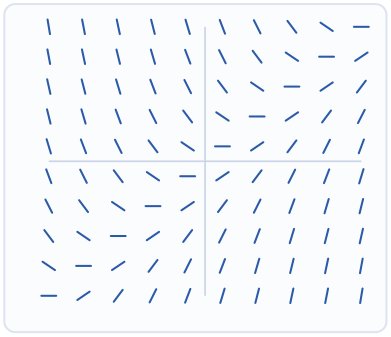
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Variation of Parameters & Greens Kernel Practice Exercises

1 MCQ

Find a particular solution of $y'' - y = 2$ using variation of parameters.



A 2

B -2

C -1

D $-\frac{1}{2}$

2 MCQ

Find a particular solution of $y'' - 4y' + 4y = 6$ using variation of parameters.

A $-\frac{3}{2}$

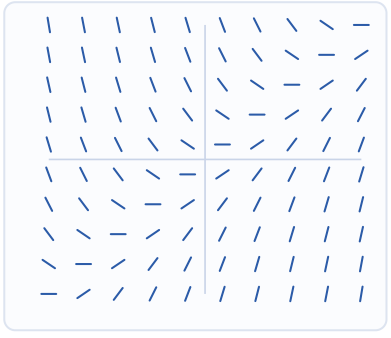
B 3

C $\frac{3}{2}$

D 6

3 MCQ

Find a particular solution of $y'' + 4y = 8$ using variation of parameters.



A 2

B -2

C 4

D 8

4 MCQ

Find a particular solution of $y'' + y = \cot x$ (for $0 < x < \pi$) using variation of parameters.

A $\sin x \ln|\csc x + \cot x|$

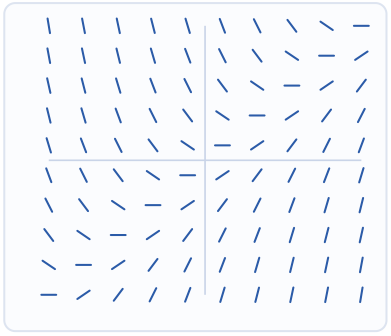
B $-\cos x \ln|\csc x + \cot x|$

C $-\sin x \ln|\sec x + \tan x|$

D $-\sin x \ln|\csc x + \cot x|$

5 MCQ

Find a particular solution of $y'' - 4y = e^{2x}$ using variation of parameters.



A $(\frac{x}{4} + \frac{1}{16})e^{2x}$

B $(\frac{x}{4} - \frac{1}{16})e^{2x}$

C $(\frac{x}{2} - \frac{1}{16})e^{2x}$

D $-(\frac{x}{4} - \frac{1}{16})e^{2x}$

6 MCQ

Find a particular solution of $y'' + 4y = \csc 2x$ (for $0 < x < \pi/2$) using variation of parameters.

A $-\frac{x}{2}\cos 2x + \frac{1}{4}\sin 2x \ln|\sin 2x|$

B $\frac{x}{2}\cos 2x + \frac{1}{4}\sin 2x \ln|\sin 2x|$

C $-\frac{x}{2}\cos 2x - \frac{1}{4}\sin 2x \ln|\sin 2x|$

D $-\frac{x}{4}\cos 2x + \frac{1}{4}\sin 2x \ln|\sin 2x|$

7 MCQ

Find a particular solution of $y'' - 6y' + 9y = \dfrac{e^{3x}}{x^2}$ (for $x > 0$) using variation of parameters.

A $e^{3x}(\ln x + 1)$

B $-e^{3x}(\ln x - 1)$

C $-e^{3x}(\ln x + 1)$

D $-e^{3x}\ln x$

8 MCQ

Find a particular solution of $y'' + 4y' + 3y = e^{-x}$ using variation of parameters.

A $(\frac{x}{2} + \frac{1}{4})e^{-x}$

B $(x - \frac{1}{4})e^{-x}$

C $-(\frac{x}{2} - \frac{1}{4})e^{-x}$

D $(\frac{x}{2} - \frac{1}{4})e^{-x}$

9 MCQ

The second-order linear homogeneous ODE $F''(w) - 50F'(w) + 621F(w) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

A 31

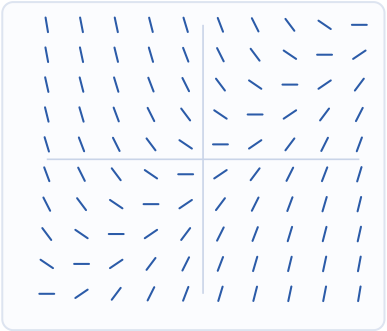
B 73

C 50

D 53

10 MCQ

Consider the second-order linear nonhomogeneous differential equation $y'' - y = \sin x$. If $y_p(x) = A\cos x + B\sin x$ is a particular solution, find the values of the constants A and B.



A $A = \frac{1}{2}, B = 0$

B $A = 0, B = \frac{1}{2}$

C $A = -\frac{1}{2}, B = 0$

D $A = 0, B = -\frac{1}{2}$

11 MCQ

Consider the second-order linear non-homogeneous differential equation: $\frac{d^2y}{dx^2} + 4y = \sec(2x)$. Suppose a particular solution is found using the method of variation of parameters. Let the fundamental set of solutions for the associated homogeneous equation be $y_1(x) = \cos(2x)$ and $y_2(x) = \sin(2x)$. If the particular solution is expressed in the form $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$, determine the value of $u_2'(x)$.

A $\frac{1}{4}$

B $\frac{1}{2}$

C $\frac{1}{2}\sin(2x)$

D $\frac{1}{4}\cos(2x)$

12 MCQ

The separable initial value problem $\frac{dF}{dw} = 12w$ satisfies $F(0) = 9$. Determine the solution value $F(12)$.

A 873

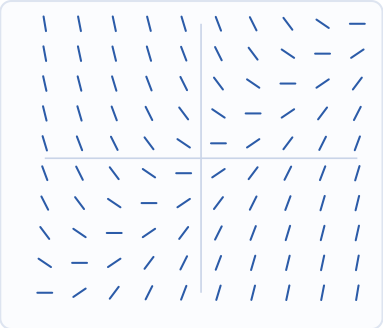
B 876

C 879

D 864

13 MCQ

Solve $y'' - 2y' + 5y = 0$, $y(0) = 2$, $y'(0) = 1$. Find $y(\pi/2)$.



A $2e^{\pi/2}$

B $-2e^{\pi}$

C $-e^{\pi/2}$

D $-2e^{\pi/2}$

14 MCQ

Find a particular solution of $y'' - 4y' + 3y = 6x + 4$.

A $2x - 4$

B $2x + 2$

C $2x + 4$

D $4x + 4$

15 MCQ

Solve $y'' - 3y' - 4y = 0$, $y(0) = -4$, $y'(0) = -1$. Find $y(1)$.

A $-3e^4 - e^{-1}$

B $-e^4 - \frac{3}{e}$

C $\frac{3}{e} + e^4$

D $-\frac{3}{e} + e^4$

16 MCQ

Obtain the value(s) satisfying $y'' - 5y' + 4y = 0$, $y(0) = 3$, $y'(0) = 6$. Find $y(1)$.

A $2e + e^4$

B $-2e + e^4$

C $e + 2e^4$

D $-e^4 - 2e$

17 MCQ

Solve $y'' - 2y' + y = 0$, $y(0) = -2$, $y'(0) = -5$. Find $y(1)$.

A $-\frac{5}{e}$

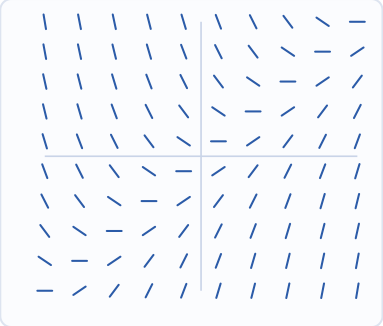
B e

C $5e$

D $-5e$

18 MCQ

Solve $y'' + y = 0, y(0) = -3, y'(0) = -3$. Find $y(\pi)$.



A $\frac{3}{2}$

B 3

C -3

D 6

19 MCQ

Solve $y'' + 9y = 0, y(0) = 3, y'(0) = 9$. Find $y(\frac{\pi}{6})$.

A -3

B 3

C 6

D $\frac{3}{2}$

20 MCQ

Find the root(s) satisfying $y'' - 7y' + 12y = 0, y(0) = 4, y'(0) = 13$. Find $y(2)$.

A $3e^6 + e^8$

B $-3e^6 + e^8$

C $e^6 + 3e^8$

D $-e^8 - 3e^6$

21 MCQ

Obtain the value(s) satisfying $y'' - 4y' + 3y = 0, y(0) = 0, y'(0) = 8$. Find $y(2)$.

A $-4e^6 + 4e^2$

B $-4e^2 + 4e^6$

C $-8e^2 + 8e^6$

D $-4e^6 - 4e^2$

22 MCQ

Calculate the solution set for $y'' - 4y' + 4y = 0, y(0) = -4, y'(0) = -11$. Find $y(2)$.

A $2e^4$

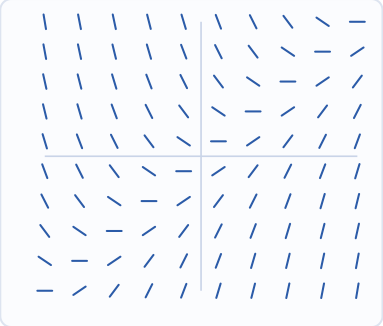
B $-11e^4$

C $10e^4$

D $-10e^4$

23 MCQ

Obtain the value(s) satisfying $y'' + 8y' + 16y = 0$, $y(0) = -2$, $y'(0) = 5$. Find $y(2)$.



A $\frac{4}{e^8}$

B $-\frac{7}{e^8}$

C $-\frac{8}{e^8}$

D $\frac{8}{e^8}$

24 MCQ

Calculate the solution set for $y'' + 4y' + 5y = 0$, $y(0) = 2$, $y'(0) = -2$. Find $y(\frac{\pi}{4})$.

A 0

B $-\frac{2\sqrt{2}}{e^{\frac{\pi}{2}}}$

C $\frac{2\sqrt{2}}{e^{\frac{\pi}{2}}}$

D $2\sqrt{2}e^{\frac{\pi}{2}}$

25 MCQ

Calculate the solution set for $y'' - 4y' - 45y = 0$, $y(0) = -7$, $y'(0) = -7$. Find $y(1)$.

A $-3e^9 + \frac{4}{e^5}$

B $-4e^9 - \frac{3}{e^5}$

C $-3e^9 - \frac{4}{e^5}$

D $\frac{4}{e^5} + 3e^9$